Marcos Morales Maldonado

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Portfolio

EDUCATION

Bachelor of Science in Mechanical Engineering — **Expected Graduation: 06/2029**
The University of Texas at El Paso (UTEP) — *GPA: 3.91/4.00 | Major GPA: 4.00/4.00*
Honors and Affiliations: Honors Program; Dean's List; SHPE//MAES (Community Engagement Committee); Theta Chi Fraternity (Secretary); American Society of Mechanical Engineers (ASME, member).
Relevant Courses: Graphic & Design Fundamentals, Engineering Design Experience, Additive Manufacturing Lab, Statics.

TECHNICAL SKILLS

Proficient: 3D modeling and engineering drawings in **SolidWorks**; **CFD** external flow simulation (SOLIDWORKS Flow Simulation); **3D printing** (FDM, PETG); Microsoft Excel (formulas, charts), Word, PowerPoint.
Working Knowledge: JavaScript, basic electrical circuits.
Languages: Spanish (native), English (professional), French (elementary), Japanese (JLPT N5).

PROJECTS

Autonomous Person-Tracking Drone - S.U.N. Program — **06/2026 - Present**
DESIGN LEAD - UTEP SHPE//MAES — *El Paso, TX*
- Designed the full ducted quadcopter frame in **SolidWorks** with **4 ducted rotors**, built on a **$600 team budget**.
- Added **12 vent slots** and side vents to cool the electronics, and balanced the internal layout for stable flight.
- Ran a **CFD study** at **15 m/s** in 2 flight cases (level and **25° tilt**), measuring **1.36 N of drag** and **0.98 N of downforce**.
- Found the duct wake as the main drag source; drag drops to **~0.15 N** at the 5 m/s search speed.
- Supported adapting **YOLO** and **MobileNet-SSD** object detection for **facial recognition** against a face database.

Orion Robotics - University Rover Challenge (URC) — **08/2026 - Present**
DESIGN & MODELING / MECHATRONICS MEMBER - University of Texas at El Paso — *El Paso, TX*
- Member of a newly formed robotics team building a rover for **URC 2027** in Utah and an autonomous **RoboRacer**.
- Assist sub-team leads with early **chassis design research** and **component selection** while learning **ROS 2** and sensor integration.

CO_2 Dragster - 1st Place out of 28 Cars — **03/2026 - 04/2026**
DESIGN TEAM PROJECT - Graphic & Design Fundamentals (MECH 1305) — *El Paso, TX*
- Won **1st place out of 28 cars** with a **0.972 s** run, **0.087 s (8%) faster** than second place.
- Compared **4 concepts**, modeled the final design in **SolidWorks**, and cut it from wood on a **band saw**.
- Ran a **CFD study** at **20 m/s**, measuring **0.11 N of drag** and finding the wake behind the flat rear as the main drag source.

Solar Flair - Solar-Powered LED Flashlight — **01/2026 - 04/2026**
PRODUCT DESIGN TEAM PROJECT - Engineering Design Experience (ENGR 1302) — *El Paso, TX*
- Defined requirements from user surveys, used **design matrices**, and designed a **3D printed PETG** body with a **5-person team**.
- Delivered **250 lumens**, up to **6 hours** of run time, a **1200 mAh** solar-charged battery, and a **0.25 kg** body.
- Passed **6 of 6 performance tests**: 1 m drop, IPX4 splash, brightness, run time, **9-hour** solar charge, and tensile strength.
- Ran usability testing with **10 users**: **90%** rated it easy to use and reliable.

Print Orientation and Anisotropy Study — **Fall 2026**
COURSE PROJECT - Additive Manufacturing Lab (MECH 2132) — *El Paso, TX*
- Printed tensile specimens at **3 orientations (0°, 45°, 90°)** and compared **ultimate tensile strength** to measure anisotropy.

LEADERSHIP EXPERIENCE

Theta Chi Fraternity — **2026 - Present**
Secretary - University of Texas at El Paso — *El Paso, TX*
- Record meeting minutes and manage chapter communications.
- Help design and oversee the chapter's **wellness, risk management, and safety** policies to protect members.

SHPE//MAES — **2026 - Present**
Athletics & Community Engagement Committee Member - University of Texas at El Paso — *El Paso, TX*
- Organize and run athletic and social events for chapter members, refereeing games and explaining the rules.
- Selected to attend the **2026 SHPE National Convention** in Indianapolis.

WORK AND VOLUNTEER EXPERIENCE

Grove — **2026 - Present**
Host and Barista — *El Paso, TX*
- Prepare drinks, take orders, and seat guests while balancing work with a full engineering course load.

Chipotle Mexican Grill — **2025 - 2026**
Grill Prep — *El Paso, TX*
- Ran the dish station solo, prepped protein for the next day, and supported the line during rush hours.

Community Volunteer — **2025 - Present**
Kelly Center for Hunger Relief; Karol G and Usher concerts — *El Paso, TX*
- Supported food relief efforts, and guided concert attendees, answered questions, and handed out water in line.